



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
PART TEST
06-12-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TOAS, TAAS, TNAS, I(A) & LIVE-I(A)] (SCORE-I)

Time : 3 Hours

PAPER-2

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

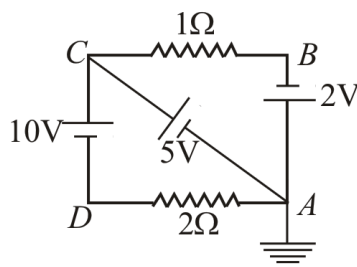
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
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Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

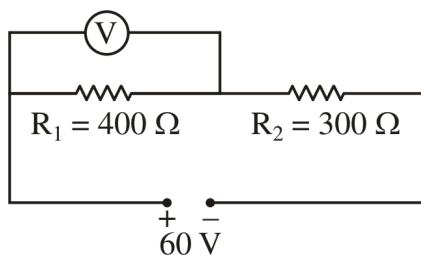
1. In the circuit shown in the figure, the value of $V_B + V_D$:-



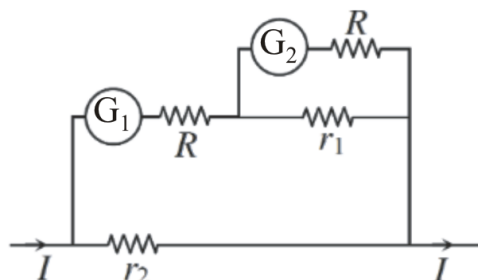
- (A) - 3V (B) 7V
(C) - 7V (D) 3V
2. Choose correct statement(s) :
- (1) The number of significant figures in 0.31×10^3 m is two.
 - (2) The length and breadth of a rectangular sheet are 4.234 m and 1.00 m then the perimeter of rectangle with correct number of significant figures is 10.46 m.
 - (3) When 97.52 m is divided by 2.54 s, the result with correct number of significant figures is 38.393.
 - (4) When 0.2 J is subtracted from 7.26 J, then result with correct number of significant figure is 7.1 J.
- (A) Only 1 and 2 are correct. (B) Only 1, 2 and 3 are correct.
(C) Only 2 and 3 are correct. (D) Only 1 and 4 are correct.

Space for Rough Work

3. In the circuit shown, the voltmeter reads 30 V when connected across the $400\ \Omega$ resistance. The same voltmeter when connected across $300\ \Omega$ will measure :



- (A) 20 V (B) 22.5 V (C) 25 V (D) 40 V
4. The diagram shows two galvanometers G_1 and G_2 . When current $I = 1\text{ A}$ both G_1 and G_2 shows full scale deflection. It is given that G_1 shows full scale deflection for 5 mA and G_2 shows full scale deflection for 2 mA. The values of r_1 and r_2 are (G_1 and G_2 are of negligible resistances)



- (A) $r_1 = \frac{2R}{9}$ and $r_2 = \frac{R}{995}$
 (B) $r_1 = \frac{2R}{3}$ and $r_2 = \frac{7R}{995}$
 (C) $r_1 = \frac{3R}{2}$ and $r_2 = \frac{7R}{995}$
 (D) $r_1 = \frac{4R}{9}$ and $r_2 = \frac{4R}{90}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
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Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

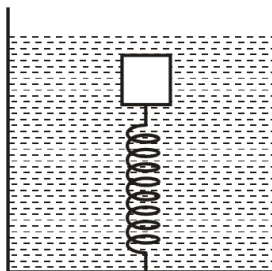
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. A block of mass ' m ' is attached by means of a spring to the bottom of a tank of water as shown in figure. At equilibrium, the spring is under compression. If the tank is now allowed to fall freely, then choose the correct alternative(s) :-



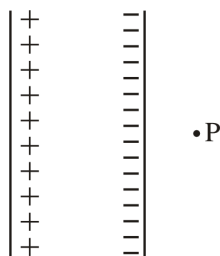
- (A) the block comes to rest when the spring is at its natural length.
- (B) the spring compresses more than its equilibrium compression.
- (C) the buoyant force becomes zero.
- (D) there will be some elongation in the spring.

Space for Rough Work

6. A piston is pressed with a force F on a hydraulic cylinder containing water ($\rho = 10^3 \text{ kg/m}^3$). The cross-sectional area of the cylinder is $A = 100 \text{ cm}^2$. The water is forced into pipe with a cross-sectional area of 1 cm^2 that rises a height $h = 50 \text{ cm}$ (Atmospheric pressure $P_0 = 10^5 \text{ N/m}^2$)



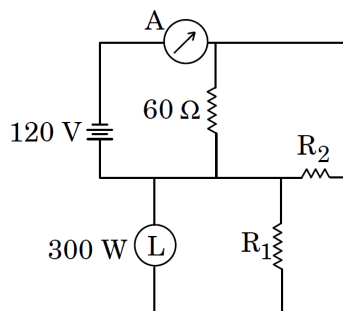
- (A) Force F required to make water eject with a speed of 10 m/s is 550 N .
 (B) Force F required to make water eject with a speed of 10 m/s at the top is 5500 N .
 (C) The pressure inside the cylinder at the bottom when water ejects at 10 m/s is $1.55 \times 10^5 \text{ Pa}$.
 (D) none of the above
7. An isolated parallel plate capacitor have unequal charges $+4Q$ and $-2Q$ on its plates. If capacitance of the capacitor is C and P is a point outside of the capacitor. Then :-



- (A) a point charge at point P will not experience any electric force due to capacitor plates.
 (B) the potential difference between the plates will be $\frac{3Q}{C}$.
 (C) the energy stored in the electric field in the region between the plates is $\frac{9Q^2}{2C}$.
 (D) the force on one plate due to the other plate is $\frac{2Q^2}{Cd}$.

Space for Rough Work

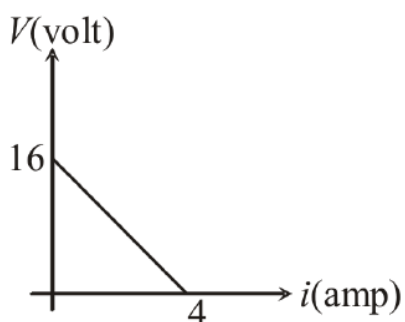
8. A cylindrical vessel C_1 of area of cross-section A and height 1 m is filled fully up to a height 0.4 m by water (density 1000 kg-m^{-3}) and remaining by liquid of density 800 kg-m^{-3} . The cylinder has small orifice at the bottom, which is closed initially. The cylinder C_1 is then immersed in another bigger cylindrical vessel C_2 of area of crosssection $6A$, containing liquid of density 800 kg-m^{-3} to the height of 0.5 m. If the orifice of C_1 is opened, then,
- (A) The speed with which the water comes out from the orifice (in ms^{-1}) will be 4
- (B) After certain interval of time, water from C_1 stops flowing out, thereby establishing the steady state of the system. The length of water column in cylinder C_1 is 6.67 cm
- (C) After certain interval of time, water from C_1 stops flowing out, thereby establishing the steady state of the system. The length of water column in cylinder C_1 is 9.5 cm
- (D) In steady state, the pressure exerted on the bottom of cylinder C_2 will be about $1.09 \times 10^5 \text{ Nm}^{-2}$ [Atmospheric pressure, $p_0 = 1 \times 10^5 \text{ Nm}^{-2}$]
9. In the circuit given below, if $R_1 = 240 \Omega$ and $R_2 = 120 \Omega$ respectively, L is a lamp of 300 W then



- (A) Power dissipated in R_1 is 60 W
- (B) Power dissipated in R_2 is 120 W
- (C) Effective resistance of the circuit is 50Ω
- (D) The ammeter A reads 6 A

Space for Rough Work

10. A battery of emf E and internal resistance r is connected across a resistance R . Resistance R can be adjusted to any value greater than or equal to zero. A graph is plotted between the current passing through the resistance (i) and potential difference across the terminals of the battery (V). Select the correct alternative (s).



- (A) Internal resistance of the battery is $4\ \Omega$.
 (B) emf of the battery is 16 V .
 (C) Maximum current which can be taken from the battery is 2 A .
 (D) Maximum thermal power dissipated across R is 16 J .

Space for Rough Work

SECTION-III : (Maximum Marks: 24)

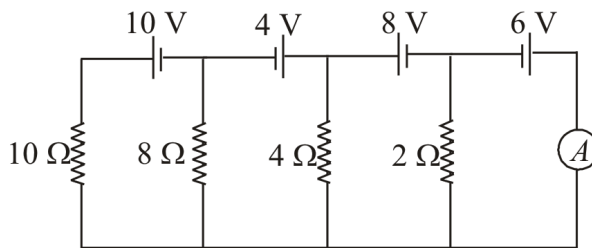
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Full Marks : +3 If only the correct answer is given.

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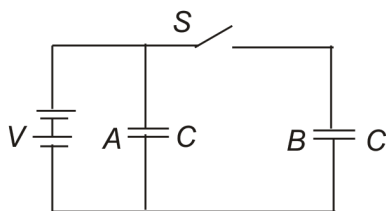
Negative Marks : -1 In all other cases

1. Find the attraction force (in Newton) between two parallel glass plates separated by a distance $h = 0.10$ mm, after a water drop of mass $m = 70$ g was introduced between them. The wetting is assumed to be complete.
 $(g = 10 \text{ m/s}^2, \rho = 10^3 \text{ kg/m}^3, T = \frac{1}{14} \times 10^{-3} \text{ N/m})$
2. The reading of the ideal ammeter connected in the given circuit is $0.39 \times N$. The Value of N is :-
 (Assume that the cells have negligible internal resistance)

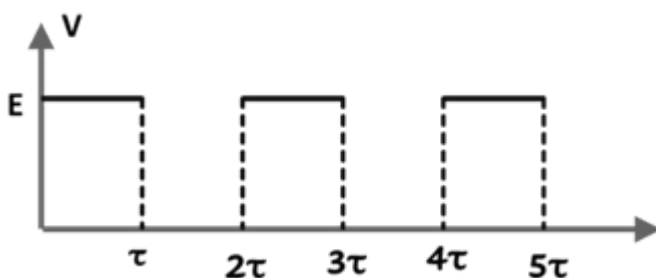


Space for Rough Work

3. The figure shows two identical parallel plate capacitors connected to a battery with switch S closed. The switch is now opened and the free space between the plates of the capacitors is filled with a dielectric of dielectric constant (or relative permittivity) 3. The ratio of the total electrostatic energy stored in both capacitors before and after the introduction of the dielectric is $3/N$. Then find the value of N.



4. A resistor of resistance R and a capacitor of capacitance C is connected in series with the voltage source. The EMF of the source changes periodically as shown in the figure. The maximum charge on the capacitor is $\frac{e}{e+N} \times CE$. The value of N is $-\tau$ (τ is time constant)

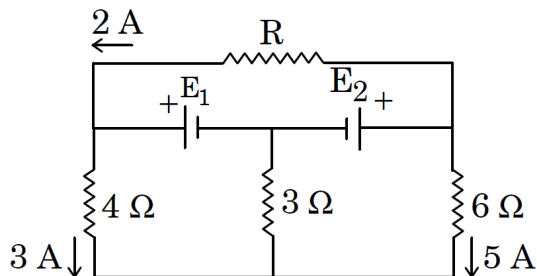


5. The space between the plates of a parallel plate capacitor is completely filled with a material of resistivity $2 \times 10^{11} \Omega\text{-m}$ and dielectric constant 6. Capacity of the capacitor with the given dielectric medium between the plates is 20 nF. Find the leakage current (in μA) if a potential difference 2500 V is applied across the capacitor. (Give your answer in nearest integer.)

Space for Rough Work

6. A tube of conical bore of length 10cm is just dipped inside the water. The diameters of upper and lower ends are 0.04 cm and 0.06 cm respectively. Find the height (in cm) to which the liquid rises in the tube. Given surface tension of water is 70 dyne cm^{-1} and angle of contact 0° . (Take : $g = 9.8 \text{ m/s}^2$) (Give your answer in nearest integer.)

7. In the circuit shown in Figure,



current in the 3Ω resistor is

8. A open organ pipe containing air resonates in fundamental mode due to a tuning fork. The measurement value of length ℓ (in cm) of the pipe are $\ell = 94.0 \pm 0.1$, $r = 5.0 \pm 0.05$. The velocity of the sound in air is accurately known. The maximum percentage error in the measurement of the frequency of that tuning fork by thus experiment, will be α , then find the value of 25α .

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

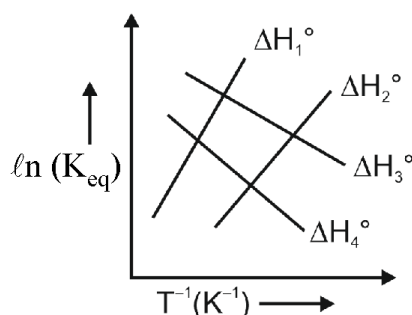
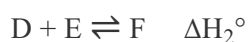
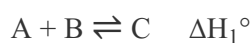
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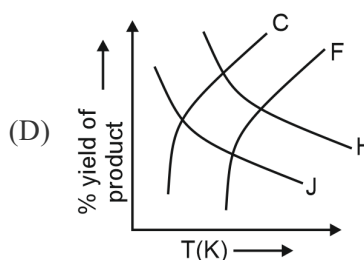
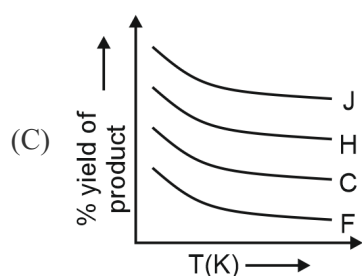
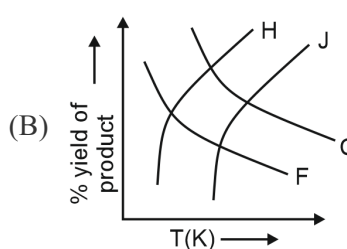
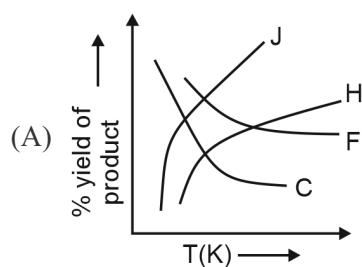
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1. Consider following four reactions and given plot showing variation of equilibrium constant (K_{eq}) with temperature :

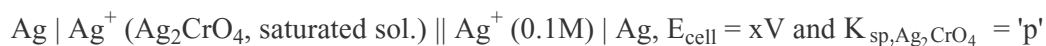


The correct graph showing the variation of % yield of product, C, F, H and J with temperature is (ΔH° is independent of temperature, $\ln K$ is the natural log of equilibrium constant)



Space for Rough Work

2. Consider the four different cells having equal cell potential is given conditions :



The correct option for solubility product for all the four cases is (by considering all cells are in equilibrium stage)

- (A) $s > q > p > r$ (B) $s < p < q < r$ (C) $r > s > q > p$ (D) $r > q > p > s$
3. 1 g of charcoal adsorbs 100 ml of 0.5 M CH_3COOH to form a monolayer, and thereby the molarity of CH_3COOH reduces to 0.49. Find the surface area of the charcoal adsorbed by each molecule of acetic acid. Surface area of charcoal = $3.01 \times 10^2 \text{ m}^2/\text{g}$.
- (A) $5 \times 10^{-19} \text{ m}^2$ (B) $9 \times 10^{-19} \text{ m}^2$ (C) $20 \times 10^{-19} \text{ m}^2$ (D) $25 \times 10^{-19} \text{ m}^2$
4. 10 g of a weak monocarboxylic acid in 100 g of benzene freezes at 3.35°C . Assume that acid undergoes complete association in benzene. Find the degree of dissociation of 10 g of acid in 100 g of water whose freezing point is found to be -1.75°C . Given: $K_f(\text{H}_2\text{O}) = 1.86 \text{ K-Kg/mole}$, $K_f(\text{benzene}) = 5.12 \text{ K-Kg/mole}$ and freezing point of benzene = 5.51°C .
- (A) 0.05 (B) 0.115 (C) 0.621 (D) 0.825

Space for Rough Work

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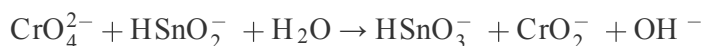
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5. The chromate ion, CrO_4^{2-} oxidizes the stannite ion, HSnO_2^- to the stannate ion HSnO_3^- and is itself reduced to chromite ion, CrO_2^- in basic medium as per the following reaction:



If 400 ml of M/5 chromate ion react with 500 ml of M/4 stannite ion then:

- (A) Chromate ion, CrO_4^{2-} is the limiting reagent
- (B) Stannite ion HSnO_2^- is the limiting reagent
- (C) At the end of reaction, the concentration of CrO_2^- ions $\approx 0.09\text{M}$
- (D) At the end of reaction, the concentration of HSnO_3^- ions $\approx 0.13\text{M}$

Space for Rough Work

6. One mole of an ideal monoatomic gas initially at a temperature T_1 , pressure P_1 and volume V_1 in a thermally insulated cylinder closed at one end and fitted with a movable piston at the other end is suddenly compressed placing a weight on the piston. The gas pressure is raised to $2P_1$. Which of the following is correct?

(A) Rise in temperature $= \Delta T = \frac{2}{5} T_1$ (B) Change in volume $= \Delta V = \frac{-3}{10} \frac{RT_1}{P_1}$

(C) Final temperature $= \frac{7}{5} T_1$ (D) $P_1 V_1^1 = (2P_1) V_2^1$

7. A sample of H_2O_2 solution labelled as 34.05 volume has density of 1.128 gm/ml. Mark the correct option(s) representing concentration of same solution in other units. [Solution contains only H_2O and H_2O_2]

(A) Mole fraction of H_2O_2 in the solution $= 0.5$

(B) %w/v $= 10.2\%$

(C) $M_{H_2O_2} = 6M$

(D) $m_{H_2O_2} = 2.924 m$

Space for Rough Work

8. For phase transition involving 3 moles of benzene -
 $C_6H_6(l, 77^\circ C, 1 \text{ atm}) \rightarrow C_6H_6(g, 77^\circ C, 0.5 \text{ atm})$; $\Delta H_{\text{vap}} = 1.4 \text{ kJ/mole}$
 If $77^\circ C$ is normal boiling point of benzene then correct value(s) of change in thermodynamic parameters for process is : [Use : $R = 8 \text{ J/K-mole}$, $\ln 2 = 0.7$]
- (A) $\Delta S = 20.8 \text{ J/K}$ (B) $\Delta G = -5.88 \text{ kJ}$ (C) $\Delta U < 0$ (D) $\Delta U > 0$
9. Which of following statement(s) is/are correct for Zeolites :
- (A) They are hydrated alumino-silicates which can be used for shape selective catalysis
- (B) They have general formula $M_{x/n}[(AlO_2)_x(SiO_2)_y].mH_2O$ where, n is the charge on the metal cation M^{n+}
- (C) They have pore sizes between 260 pm to 740 pm
- (D) ZSM-5 which is used in petroleum industry as a catalyst, converts hydrocarbons into alcohols is an example of Zeolites.
10. Choose the correct statement(s) among the following -
- (A) Internal energy of H_2O remains constant during conversion of liquid into its vapour at constant temperature
- (B) During fusion of ice into water, enthalpy change and internal energy change are almost same at constant temperature
- (C) Molar heat capacity of diatomic gases are temperature dependent
- (D) During boiling of H_2O at 1 atm, $100^\circ C$, average kinetic energy of H_2O molecules increases

Space for Rough Work

SECTION-III : (Maximum Marks: 24)

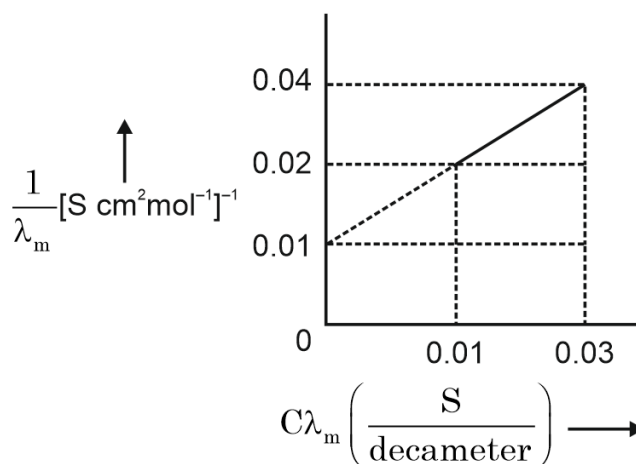
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1. The variation of $\frac{1}{\lambda_m}$ vs $C \lambda_m$ for RCOOH at 298 K has been shown in the graph. Find pK_a of RCOOH.



2. A 150 ml solution of I_2 is divided into two unequal parts. First part reacts with hypo solution in acidic medium. 15 ml of 0.4 M hypo was consumed. Second part was treated with 100 ml of 0.3M NaOH solution yielding NaI and $NaIO_3$. Residual base required 10 ml of 0.3M H_2SO_4 solution for complete neutralization. Calculate the value of "10 times" the initial concentration of I_2 ?

Space for Rough Work

3. An aromatic compound contains 69.4% carbon and 5.8% hydrogen. A sample of 0.303 g of this compound was analyzed for nitrogen by kjeldahl's method. The ammonia evolved was absorbed in 50 ml of 0.05M H_2SO_4 . The excess of the acid required 25 ml of 0.1 M NaOH for neutralization. Its molecular mass is 121. Determine the molecular formula and express the answer in the form of $\frac{w+x-y-z}{2}$ if the molecular formula of the compound is $\text{C}_w\text{H}_x\text{N}_y\text{O}_z$.
4. 0.00012% MgSO_4 and 0.000111% CaCl_2 is present in water. What is the measured hardness of water in ppm of CaCO_3 . [Atomic mass : Mg = 24, S = 32, Ca = 40, Cl = 35.5]
5. When Aluminium pieces are added to aqueous solution of Tetrathionic acid ($\text{H}_2\text{S}_4\text{O}_6$) a pungent gas is produced. Find number of moles of Tetrathionic acid consumed per mole of Aluminium in this reaction ?
6. Calculate the quantity (in gram) of CO_2 in 550 mL of soda water when packed under 2.5 Bars CO_2 pressure at 298 K (K_H for $\text{CO}_2 = 1.67 \times 10^8 \text{ Pa}$ at 298 K)
7. A 10 ml sample of hydroxylamine solution was diluted to one litre. 50 mL of this diluted solution was boiled with excess of Fe (III) solution and the resulting solution required 10 mL of 0.02M KMnO_4 solution for complete oxidation of Fe(II). Calculate the mass (in g) hydroxylamine in 91 mL of the original hydroxylamine solution.
Balanced reaction is, $2\text{NH}_2\text{OH} + 4\text{Fe}^{3+} \rightarrow \text{N}_2\text{O}(\text{g}) + \text{H}_2\text{O} + 4\text{Fe}^{2+} + 4\text{H}^+$
8. In the following reactions:

$$\text{CaOCl}_2 \xrightarrow{\Delta} a \text{Ca}(\text{ClO}_3)_2 + b \text{CaCl}_2$$

$$\text{NaCl} + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + c \text{HCl}$$

$$\text{Na}_2\text{S}_2\text{O}_3 + d \text{HCl} \rightarrow e \text{NaCl} + \text{H}_2\text{O} + \text{S} \downarrow + \text{SO}_2$$
 What will be the value of $(a + b + c) - (d + e)$ in the balanced chemical reactions?

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. The value(s) of the real parameter 'a' for which the inequality $x^6 - 6x^5 + 12x^4 + ax^3 + 12x^2 - 6x + 1 \geq 0$ is satisfied for all real x
 (A) $[-12, 37]$ (B) $[-12, 38]$ (C) $[-17, 14]$ (D) $[-17, 35]$
2. Let $f(x) = \sin^3 x + \lambda \sin^2 x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$ and $\lambda > 0$. The interval in which λ should lie in order that $f(x)$ has exactly one minimum and one maximum is
 (A) $0 < \lambda < \frac{3}{2}$ (B) $0 < \lambda < 1$ (C) $1 < \lambda < 2$ (D) $1 < \lambda < \frac{3}{2}$
3. The value of $\lim_{x \rightarrow 2} \frac{\sqrt{(x+7)} - 3\sqrt{(2x-3)}}{\sqrt[3]{(x+6)} - 2\sqrt[3]{(3x-5)}}$ is equal to
 (A) $\frac{34}{23}$ (B) $-\frac{34}{23}$ (C) $\frac{23}{34}$ (D) 0

Space for Rough Work

4. The function defined as $f(x) = \lim_{n \rightarrow \infty} \begin{cases} \cos^{2n} x & \text{if } x < 0 \\ \sqrt[n]{1+x^n} & \text{if } 0 \leq x \leq 1 \\ \frac{1}{1+x^n} & \text{if } x > 1 \end{cases}$ which of the following does hold good.

(A) Continuous at $x = 0$, but discontinuous at $x = 1$

(B) Continuous at $x = 1$, but discontinuous at $x = 0$

(C) Continuous at both $x = 1$ and $x = 0$

(D) Discontinuous at both $x = 1$ and $x = 0$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. A function $f(x)$ is defined as
$$f(x) = \begin{cases} 6x - 5 - x^2 & x \leq 3 \\ 24x - 32 - 4x^2 & x > 3 \end{cases}$$

Tangents are made on $f(x)$ at (x, y) where $x \geq 0$ & $y \geq 0$. Let tangents $T_1 : y = m_1x + b_1$ and $T_2 : y = m_2x + b_2$ have highest and lowest y intercepts of all tangents made in first quadrant then

(A) $b_1 + b_2 = 36$

(B) $b_1 + b_2 = 28$

(C) $m_1 + m_2 = -4$

(D) Area made by y axis, $T_1 = 0$ and $T_2 = 0$ is 54 square units

Space for Rough Work

6. Suppose that $f(x)$ is defined and positive for all real numbers x . If $f(x)$ increases for $x < 0$ and decreases for all $x > 0$, and $g(x) = \frac{1}{f\left(\frac{1}{x}\right)}$, then
- (A) g increases for all real x (B) g increases for $x < 0$
 (C) g decreases for $x > 0$ (D) g increases for $x > 0$
7. Let $f(x)$ be double differentiable function such that $|f'(x)| \leq 5 \forall x \in [0, 4]$ and f takes its largest value at an interior point of this interval. Then the value of $|f'(0)| + |f'(4)|$ can be
- (A) 18 (B) 19 (C) 20 (D) 21
8. If $F(x) = f(x)g(x)$ and $f'(x)g'(x) = c$, then
- (A) $F' = c \left[\frac{f}{f'} + \frac{g}{g'} \right]$ (B) $\frac{F''}{F} = \frac{f''}{f} + \frac{g''}{g} + \frac{2c}{fg}$
 (C) $\frac{F'''}{F} = \frac{f'''}{f} + \frac{g'''}{g}$ (D) $\frac{F'''}{F} = \frac{f'''}{f''} + \frac{g'''}{g''}$

Space for Rough Work

9. Let $f: [0, 1] \rightarrow \mathbb{R}$ be a double differentiable function with $f(0) = f(1) = 0$, and $f''(x) + 2f'(x) + f(x) \geq 0 \forall x \in (0, 1)$, then
- (A) $f(x) \leq 0 \forall x \in [0, 1]$
- (B) If $e^x f(x)$ assumes its minimum value in $[0, 1]$ at $x = \frac{1}{\sqrt{2}}$, then $f(x) + f'(x) < 0 \forall x \in \left(0, \frac{1}{\sqrt{2}}\right)$
- (C) If $g(x) = e^x f(x)$, then number of real solution of the equation $g(g(g(x))) = 0$ is 2
- (D) $f(x) > 0 \forall x \in [0, 1]$
10. Which of the following are true for $\forall x \in (0, \infty)$
- (A) $\ln(1+x) > x - \frac{x^2}{2}$
- (B) $\ln(1+x) > \frac{x}{x+1}$
- (C) $4\cos x + x > 0$
- (D) $2\tan^{-1} x < x + 1$

Space for Rough Work

SECTION-III : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. For the function $f(x) = \lim_{n \rightarrow \infty} \frac{\ln([1 + 9x^{2n}]) - 9^n x^{2n} \cos(9x^2)}{1 + 9^n x^{2n}}$ number of points of discontinuity in $(-3, 3)$ is: (where $[.]$ represents greatest integer function)
2. Let C be the curve $f(x) = (\ln x)^2 + 2 \ln x$ and $A(a, f(a))$, $B(b, f(b))$ where $(a < b)$ are the points of tangency of two tangents drawn from origin to the curve C. The value of the product ab is equal to
3. The smallest integral value of P for which the function $f(x) = 6Px - P \sin 4x - 5x - \sin 3x$ is monotonically increasing and has no critical point on R is:
4. Let $f(x)$ be a real valued function such that $f(x) = \frac{2x-1}{x-2} \forall x \in (2, \infty)$ and $g(x) = \frac{x^2+1}{x} + \frac{(f(x))^2+1}{f(x)}$ $\forall x > 2$, then minimum value of $g(x)$ is _____.
5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function such that $\lim_{x \rightarrow \infty} f(x) = a$ ($a \neq 0$), then $\lim_{x \rightarrow \infty} x^2 f''(x)$ is equal to

Space for Rough Work

6. A continuous and differentiable function $y = f(x)$ is such that its graph cuts line $y = mx + c$ at 11 distinct points. Then the minimum number of points at which $f'(x) = 0$ is
7. The set of all points where $f(x)$ is increasing in $(a, b) \cup (c, \infty)$ then $[a + b + c]$ is (where $[.]$ denotes the greatest integer function). Given that $f(x) = 2f\left(\frac{x^2}{2}\right) + f(6 - x^2) \forall x \in \mathbb{R}$ and $f'(x) > 0 \forall x \in \mathbb{R}$
8. The equation $k + x^3 e^{x+3} = 0$ has 2 distinct real roots and k is a prime natural number, then the number of possible value of k is
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Space for Rough Work

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work